

Chapter 10:

Analysing data

Drill practice and mixed revision of all 8 lessons · 67 questions

NSW syllabus outcome

Stage 4 Statistics: MA4-DAT-C-01 classifies and displays data using a variety of graphical representations; MA4-DAT-C-02 analyses simple datasets using measures of centre, range and shape of the data.

CHAPTER

10



Homework. Part 1 is **drill practice**: quick questions on the main skill of each lesson in Chapter 10. Write each answer on its line. Part 2 is **mixed revision**, from **Easy** to **Challenging**. Each question is tagged with its lesson, so you can look back at that lesson's summary if you get stuck.

YOU DO On your own: Part 1, drill practice.

QUESTIONS 1–30

10.01 Interpreting graphs**Easy** Reading scales: a column ends exactly halfway between two gridlines. What value does it show?

- ① between 15 and 20 ② between 45 and 50 ③ between 180 and 200 ④ between 20 and 25

Medium A survey result: write it as a percentage.

- ⑤ 5 of 50 students ⑥ 1 of 40 students ⑦ 36 of 40 students ⑧ 2 of 20 students

10.02 Misleading graphs**Easy** Misleading (M) or fair (F)?

- ⑨ The columns are different widths. ⑩ The graph is drawn in 3D, so the front column looks bigger.

Medium Explain why each graph could mislead a reader.

- ⑪ A line graph squashes the time axis to make a rise look steep.

10.03 Dot plots**Easy** Find the mode (the most common value).

- ⑫ 9, 6, 3, 7, 7, 6 ⑬ 8, 2, 6, 1, 1, 2, 2 ⑭ 2, 6, 8, 6, 7, 6, 3

Medium Find the range (largest – smallest).

- ⑮ 6, 17, 2, 6, 4, 22, 3 ⑯ 19, 22, 11, 11, 22, 8, 27 ⑰ 13, 21, 10, 37, 11, 11, 28

10.04 Stem-and-leaf plots**Easy** Write the value shown (key $3 \mid 4 = 34$).

- ⑱ $9 \mid 4$ ⑲ $8 \mid 8$ ⑳ $8 \mid 1$ ㉑ $5 \mid 1$

Medium Write the values in the row (key $3 \mid 4 = 34$).

- ㉒ $4 \mid 4\ 6\ 6\ 7$ ㉓ $4 \mid 1\ 1\ 3\ 8$ ㉔ $5 \mid 5\ 5\ 7\ 9$

10.05 Mean and mode**Easy** Find the mean: add the values, then divide by how many there are.

- ㉕ 14, 2, 8, 8 ㉖ 14, 11, 2 ㉗ 17, 13, 16, 19, 10

Medium Find the mean to 1 decimal place.

- ㉘ 39, 26, 50, 42, 7, 34 ㉙ 11, 16, 18, 31, 50 ㉚ 31, 44, 11, 39, 29

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 31–47

10.06 Median and range

Easy Find the median: order the values, then find the middle one.

31 11, 29, 5, 7, 26

32 13, 7, 18, 24, 17

33 26, 10, 28, 25, 18

Medium Find the median. There is an even number of values: take the middle two and halve their sum.

34 17, 18, 19, 10

35 15, 4, 3, 25, 38, 9

36 37, 17, 18, 40, 21, 6

10.07 Analysing dot plots and stem-and-leaf plots

Easy Find the range of the stem-and-leaf plot.

37 1 | 3 4
2 |
3 | 6
4 | 0 1 9

38 1 | 0
2 | 0 3
3 | 5
4 |
5 | 4
6 | 7

39 3 | 2
4 | 0 2
5 | 0 3
6 | 8

Medium Find the median of the stem-and-leaf plot.

40 1 | 6 7
2 |
3 | 0 3 7
4 | 1
5 | 4

41 1 | 4
2 | 8
3 | 1 6
4 | 6 6
5 | 6

42 1 | 4 7 9
2 | 3 7
3 | 0 5

10.08 Comparing data

Easy Which group is more consistent (smaller range)?

43 A: range 8, B: range 23

44 A: range 16, B: range 20

45 A: range 26, B: range 5

Medium Find the median of each group. Which median is higher?

46 A: 30, 14, 12, 37, 44
B: 21, 21, 26, 53, 28

47 A: 51, 31, 40, 13, 12
B: 40, 43, 15, 24, 16

YOU DO On your own: Part 2, mixed revision.

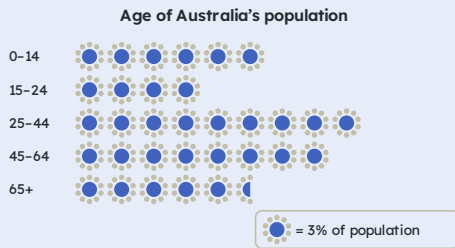
QUESTIONS 48–57

48

Easy

Use the picture graph.

Why might 25–44 be the largest group? Lesson 10.01



50

Easy

Find the range.

23, 17, 31, 29 Lesson 10.07

49

Easy

Which group is more consistent?

A: range 18, B: range 7 Lesson 10.08

51

Easy

Find the median.

8, 2, 6, 4 Lesson 10.06

52

Easy

Find the mean.

7, 8, 9, 12 Lesson 10.05

53

Easy

Write the value (key $3 \div 4 = 34$).

4 | 1 Lesson 10.04

54

Easy

Find the mode.

4, 5, 5, 5, 6, 8 Lesson 10.03

55

Easy

Is this graph misleading (M) or fair (F)?

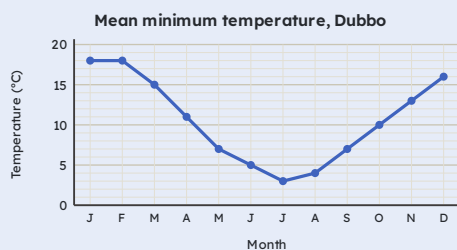
The axis starts at 90 Lesson 10.02

56

Medium

Use the line graph.

Describe the trend from January to July. Lesson 10.01



57

Medium

Group A range 12, group B range 30. Which is more consistent? Lesson 10.08

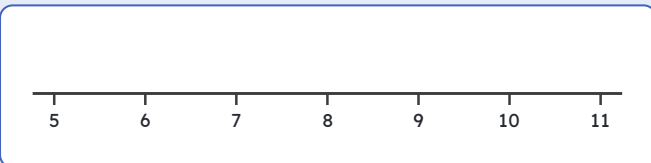
YOU DO On your own: Part 2, mixed revision.

QUESTIONS 58–67

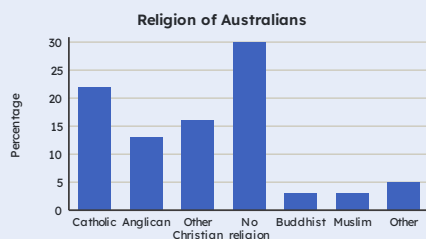
- 58** **Medium** Find the median of 5 | 1 3 4 7 (key 5 | 1 = 51). Lesson 10.07

- 60** **Medium** Find the mean: 4, 6, 8, 10, 12. Lesson 10.05

- 62** **Medium** Draw a dot plot for each set of data.
Shoe sizes: 7, 8, 8, 9, 7, 10, 8, 9, 8, 6, 8, 9 Lesson 10.03



- 64** **Challenging** Use the column graph.
How might this graph have been different 20 years ago? Lesson 10.01



- 66** **Challenging** Find the mean (2 d.p.): 3, 7, 7, 9. Lesson 10.05

- 59** **Medium** Find the range: 18, 5, 12, 30. Lesson 10.06

- 61** **Medium** Write the values in the row 4 | 0 2 7. Lesson 10.04

- 63** **Medium** A column graph's axis starts at 90. Why is that a problem? Lesson 10.02

- 65** **Challenging** Find the range of 2 | 3 8, 3 | 0 5, 4 | 1 (key 2 | 3 = 23). Lesson 10.07

- 67** **Challenging** Draw a dot plot for 1, 2, 2, 3, 3, 3, 8. Which value is the outlier? Lesson 10.03

YOU DO

On your own: extra working space. Write the question number next to your working.